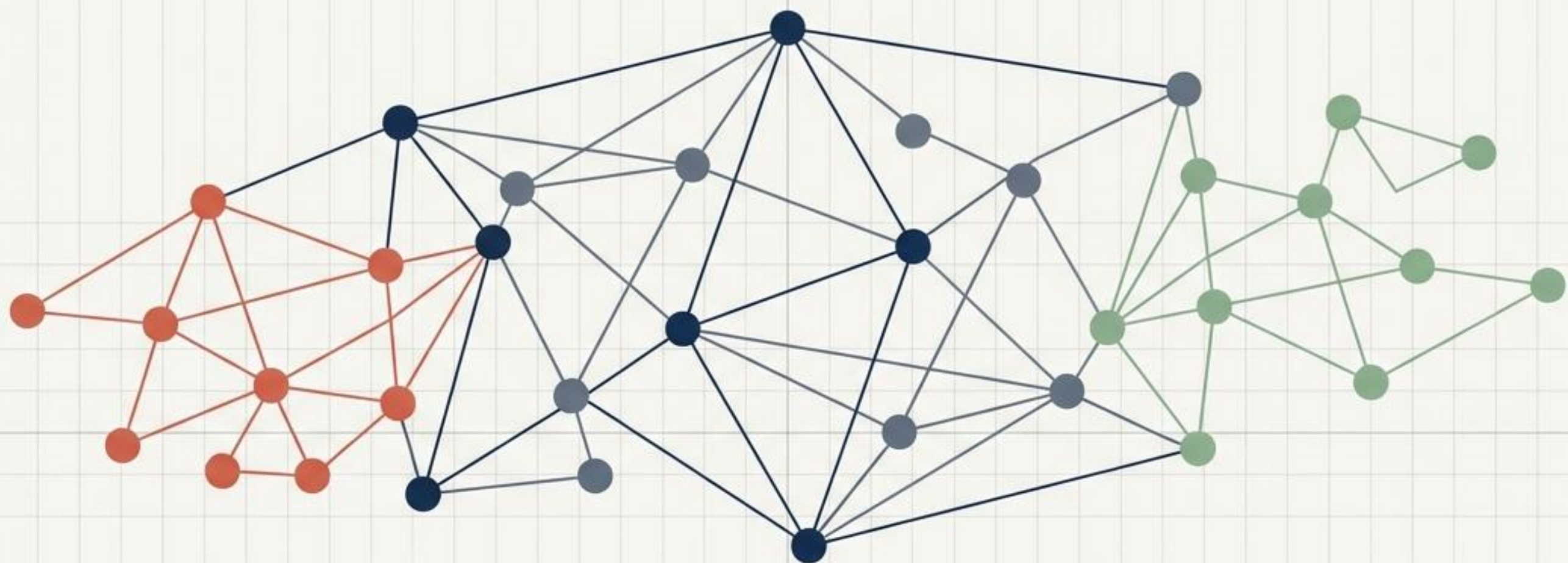




Pattern Detection Knowledge Graphs

Uncovering the Hidden Structures of Business Reality

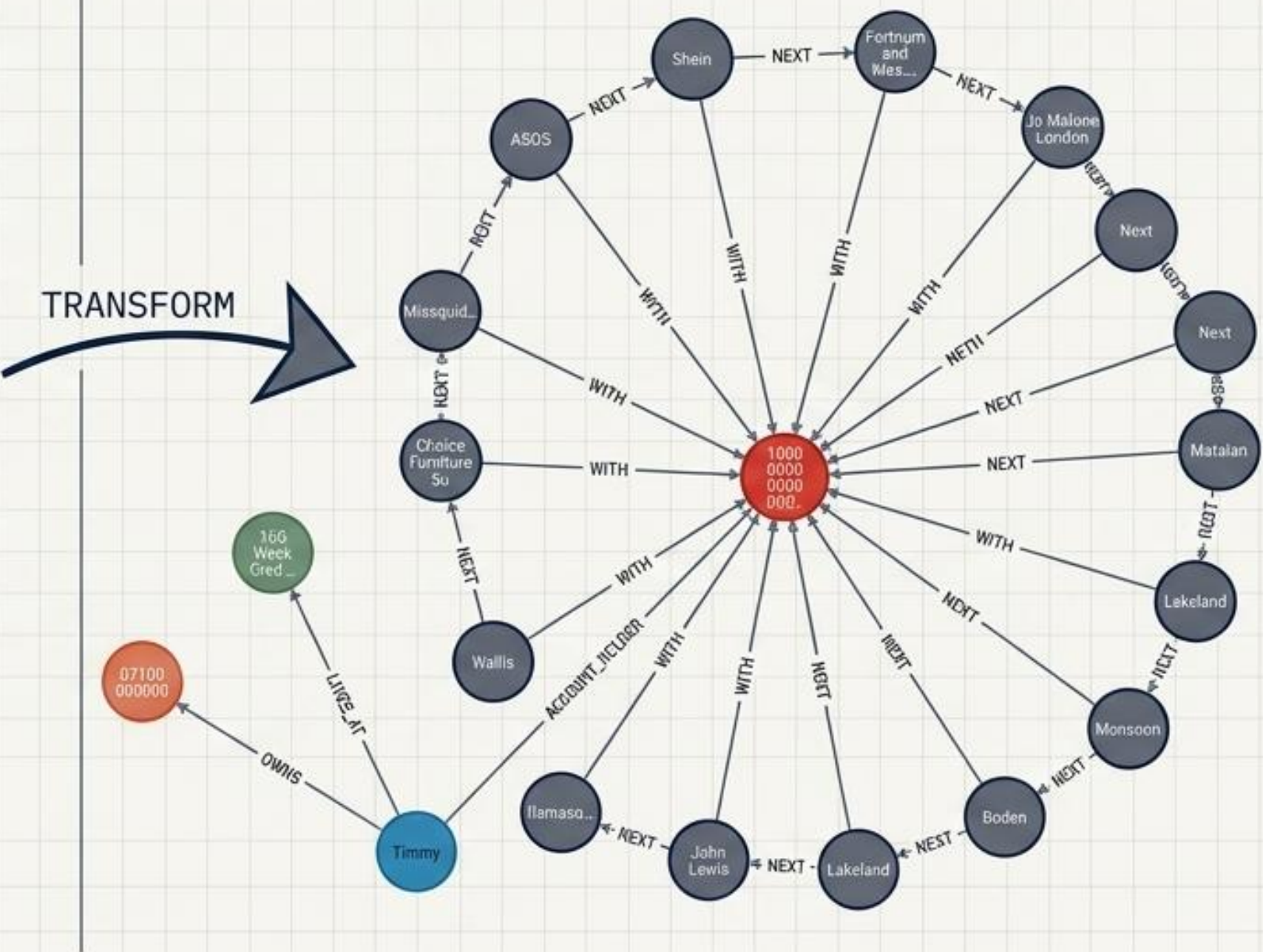
Author: dr Zbigniew Galar

The Core Premise: Changing the Data Model Changes Reality

Tabular data obfuscates structure.

"person"	"card"	"purchase"
<code>{"firstname":"Timmy","id":"231357","lastname":"Sewell"}</code>	<code>{"number":"1000 0000 0000 0000"}</code>	<code>{"date":"2022-09-02","amount":"71.60","declined":true,"retailer":"Lakeland","id":"ccffc09c-1d79-4e43-9fa9-7699bab21833"}</code>
<code>{"firstname":"Timmy","id":"231379","lastname":"Sewell"}</code>	<code>{"number":"1000 0000 0000 0000"}</code>	<code>{"date":"2022-09-04","amount":"123.82","declined":"false","retailer":"Monsoon","id":"8c61ee5b-1d8f-4da4-a928-52ddc7295203"}</code>
<code>{"firstname":"Timmy","id":"231379","lastname":"Sewell"}</code>	<code>{"number":"1000 0000 0000 0000"}</code>	<code>{"date":"2022-09-22","amount":"125.08","declined":"false","retailer":"Boden","id":"33542ef5-01bd-43d8-9d92-6c469003f0bf"}</code>
<code>{"firstname":"Timmy","id":"231379","lastname":"Sewell"}</code>	<code>{"number":"1000 0000 0000 0000"}</code>	<code>{"date":"2022-10-27","amount":"270.40","declined":"false","retailer":"Lakeland","id":"934cf536-a244-45de-9921-06686a2837a5"}</code>

Graph data illuminates context.



The Power of Pattern Detection

Centralized

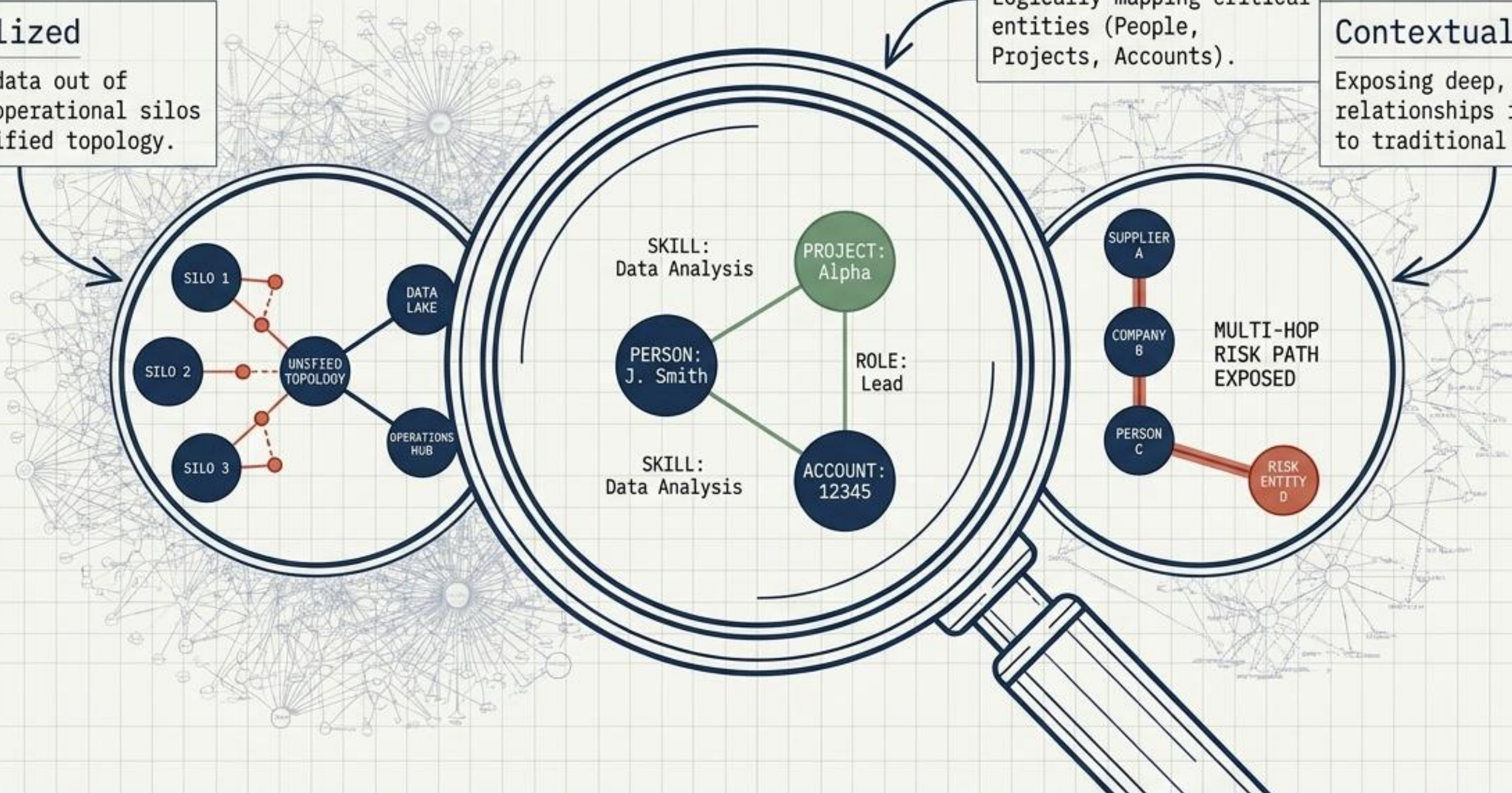
Breaking data out of
isolated operational silos
into a unified topology.

Curated

Logically mapping critical entities (People, Projects, Accounts).

Contextualized

Exposing deep, multi-hop relationships invisible to traditional queries.



Moving from rigid, slow, rule-based systems to dynamic, context-aware pattern detection.

Domain 1: Fraud Detection

\$5.8 Billion

Total cost to the US economy in 2021
(\$1B in banking alone).

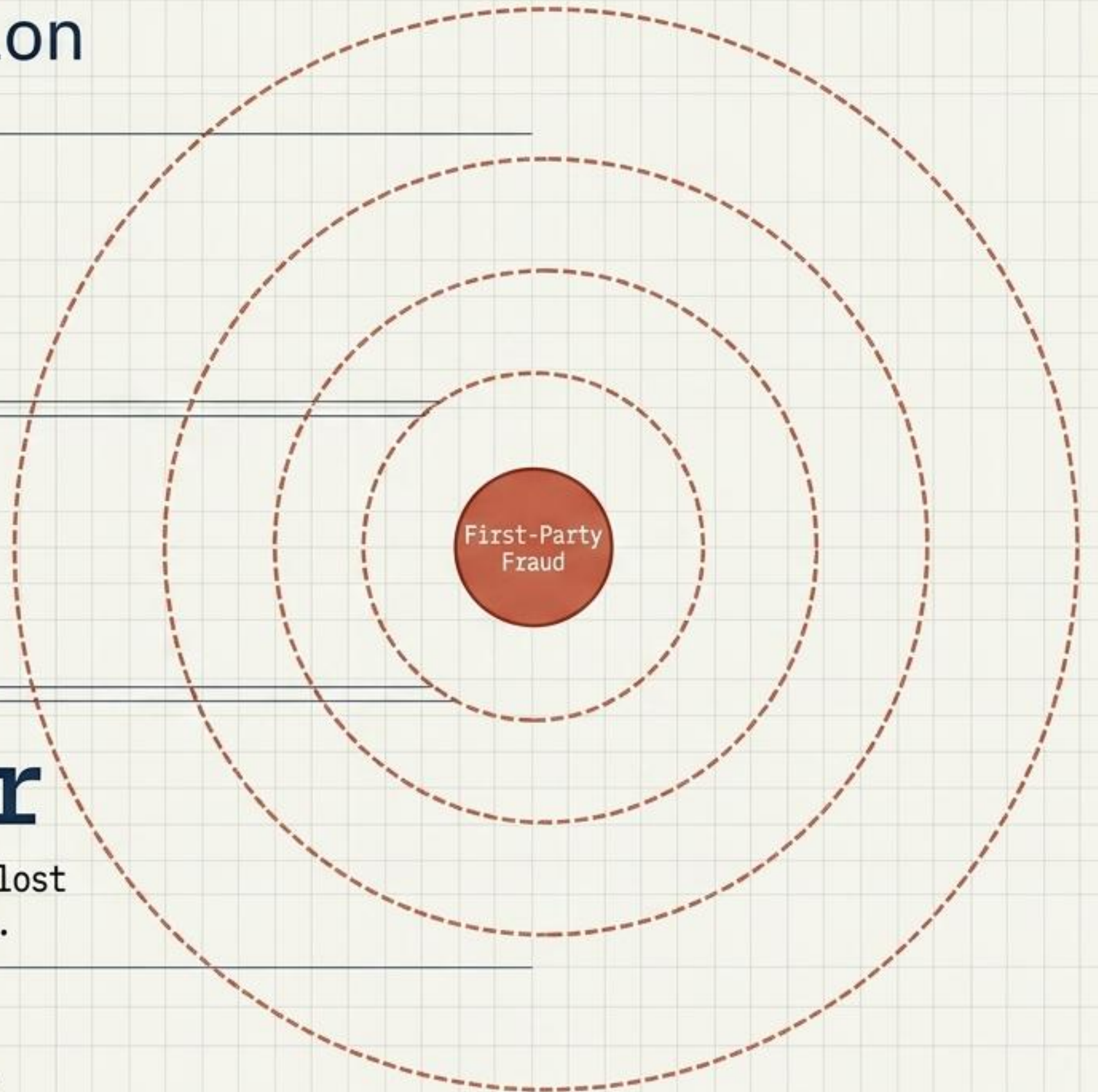
+70%

Staggering year-over-year increase in systemic fraud losses.

4.23x Multiplier

For every \$1 directly stolen, an estimated \$4.23 is lost in defensive labor, liability, and reputation damage.

Traditional defense is a losing arms race. Even marginal detection improvements yield massive ROI.



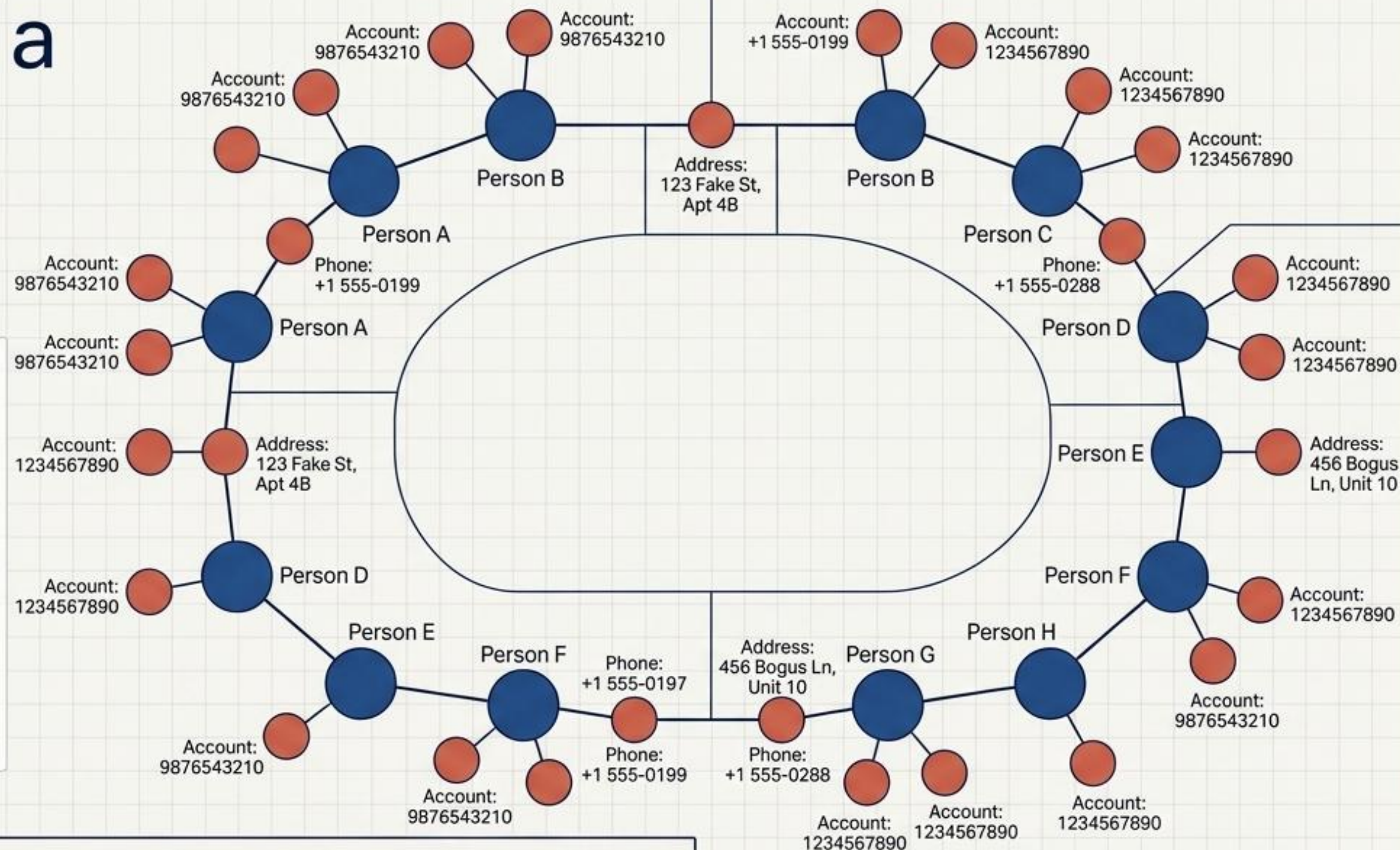
Anatomy of a Synthetic Fraud Ring

The Mechanics of First-Party Fraud

Fraudsters build substantial aggregate credit lines across multiple fake identities in good faith. When credit limits are maximized, they execute a synchronized cash-out, abandoning the accounts simultaneously.

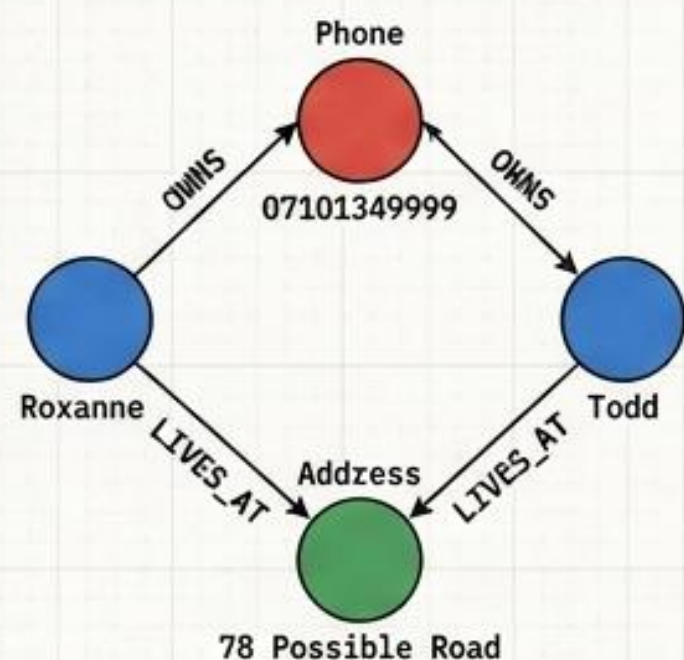
PATTERN DETECTED: Person → Shared Phone → Person → Shared Address → Person.

CONCLUSION: In a graph, the criminal network cannot hide.



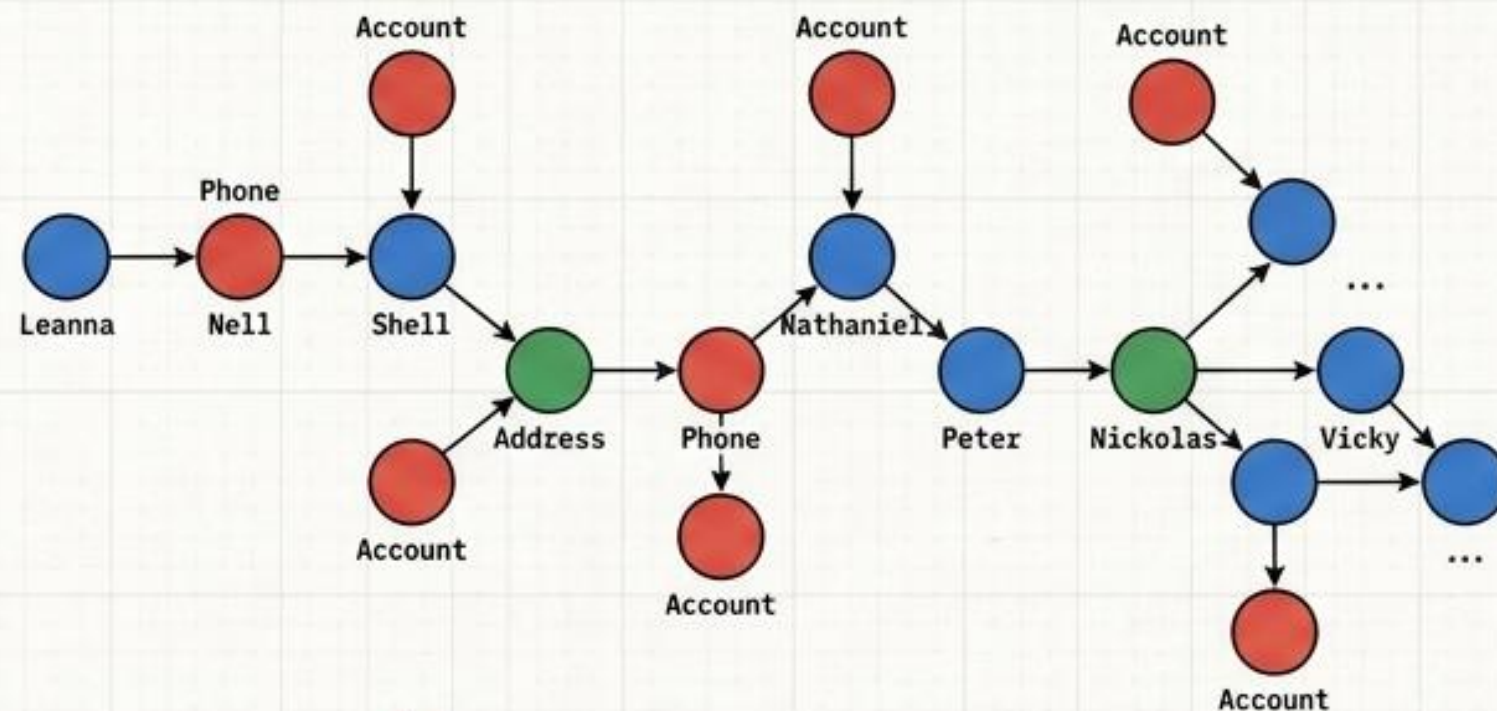
Innocent Bystander vs. Fraud Ring

Legitimate Household



SAFE PATTERN: High node overlap, localized cluster.

Synthetic Identity Ring



RISK PATTERN: Chained synthetic anchors.

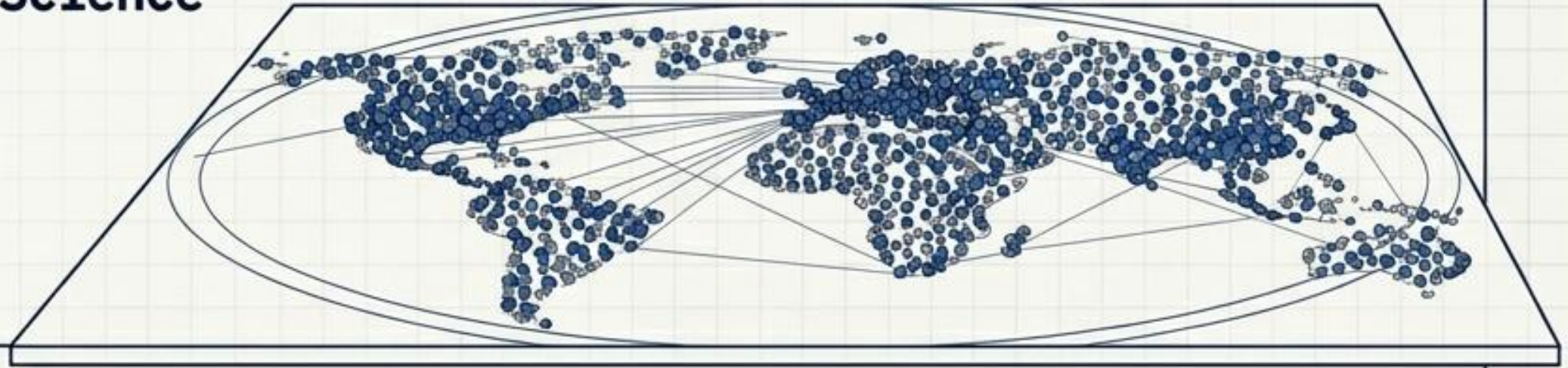
Insight: The graph topology instantly differentiates a family from a fraud ring without requiring analysts to read complex query code.

Operationalizing Defense: Speed vs. Scale

Layer 2: Global Graph Data Science

Macro-Pattern Mining

Uses deep traversal algorithms across the global graph to find newly forming rings before they execute a cash-out.



LATENCY: Periodic Batch (Hourly/Daily)

Feeds newly discovered patterns into real-time rules engine

Layer 1: Real-Time Operations

Subgraph-Local Queries

Checks immediate connections as a purchase occurs. Blocks known fraudulent topologies instantly.



LATENCY: <10 Milliseconds

Case Study: Defending Banking Circle

1. Graph Topologies

Accounts and payments are unified into a central Knowledge Graph, replacing siloed relational tables.

2. Community Detection

The Louvain algorithm systematically scans the graph to highlight abnormally dense clusters of shared identifiers.

3. ML Feature Engineering

Graph metrics feed directly into the predictive ML pipeline (e.g., risk scores of near-neighbors, topological distance to known tax havens).

Business Impact

Banking Circle transformed a slow, manual review process into a highly scalable, flexible defense system that evolves alongside sophisticated criminal tactics.

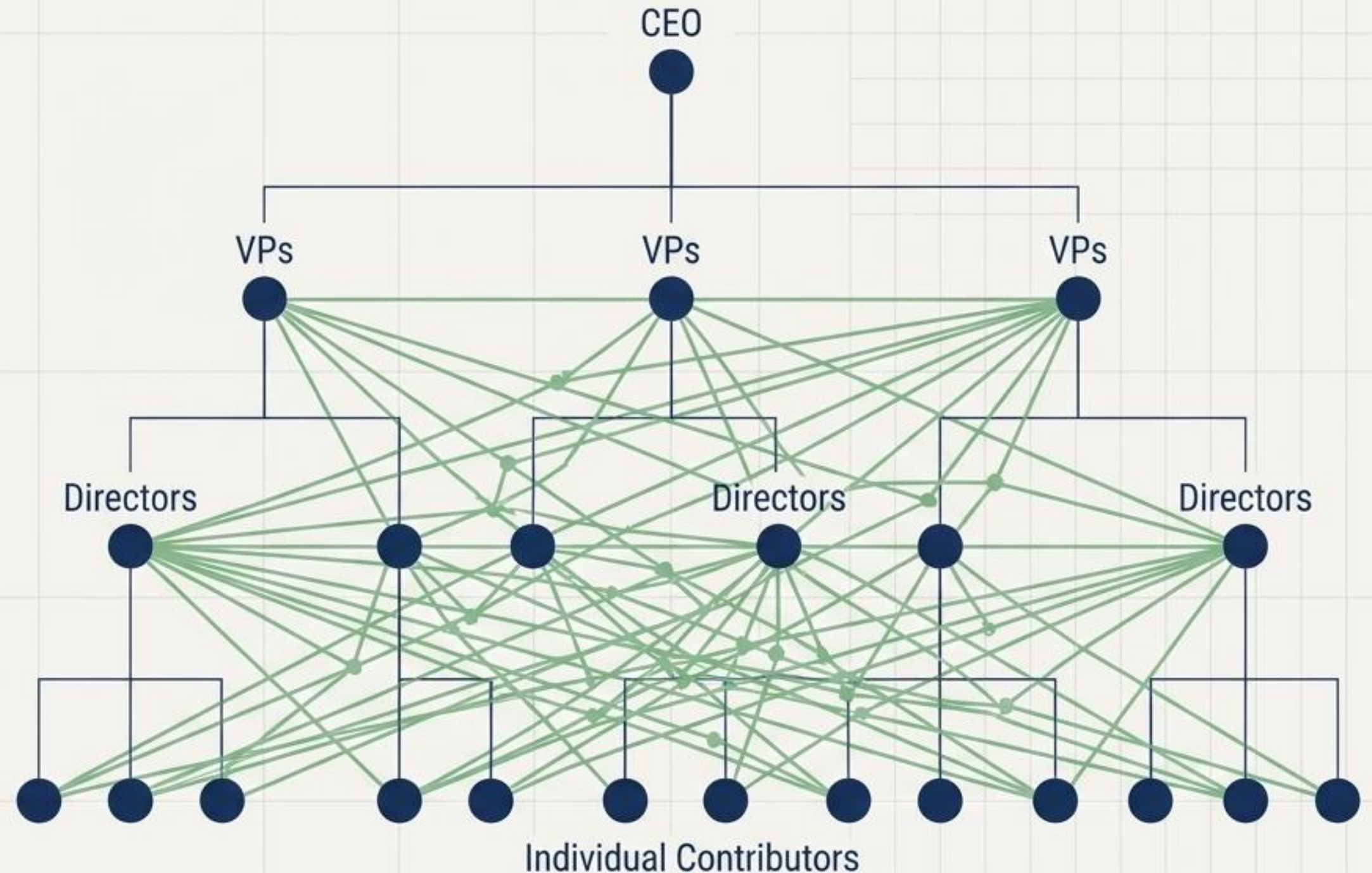
Domain 2: Skills Matching

The Illusion: The Intended Org Chart

The official hierarchical tree is merely a rigid approximation of business structure, mapping reporting lines but missing execution dynamics.

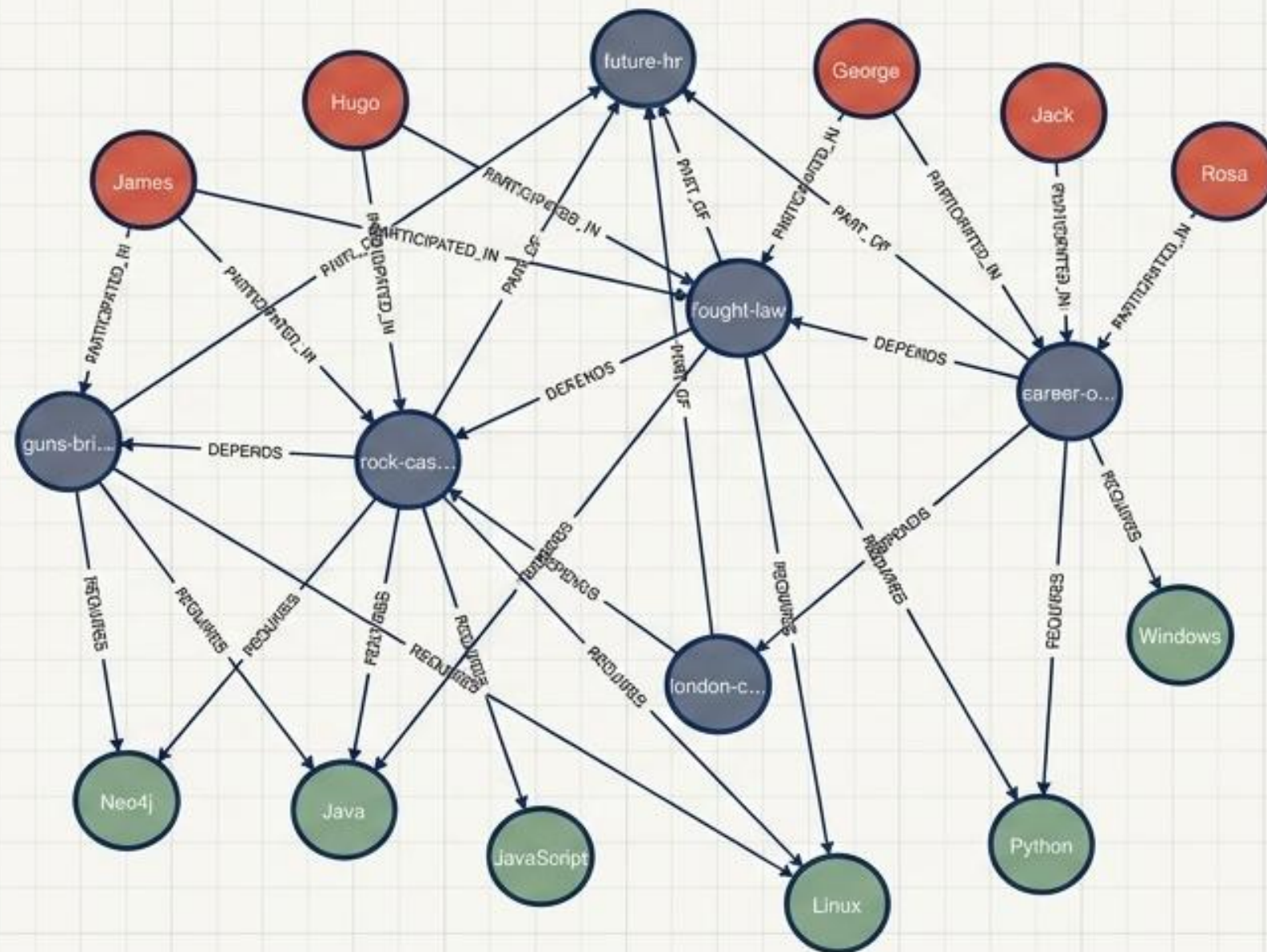
The Reality: The Collaborative Network

The informal network—driven by shared projects, actual skills, and lateral communication—dictates how work truly gets done.



Mapping True Expertise

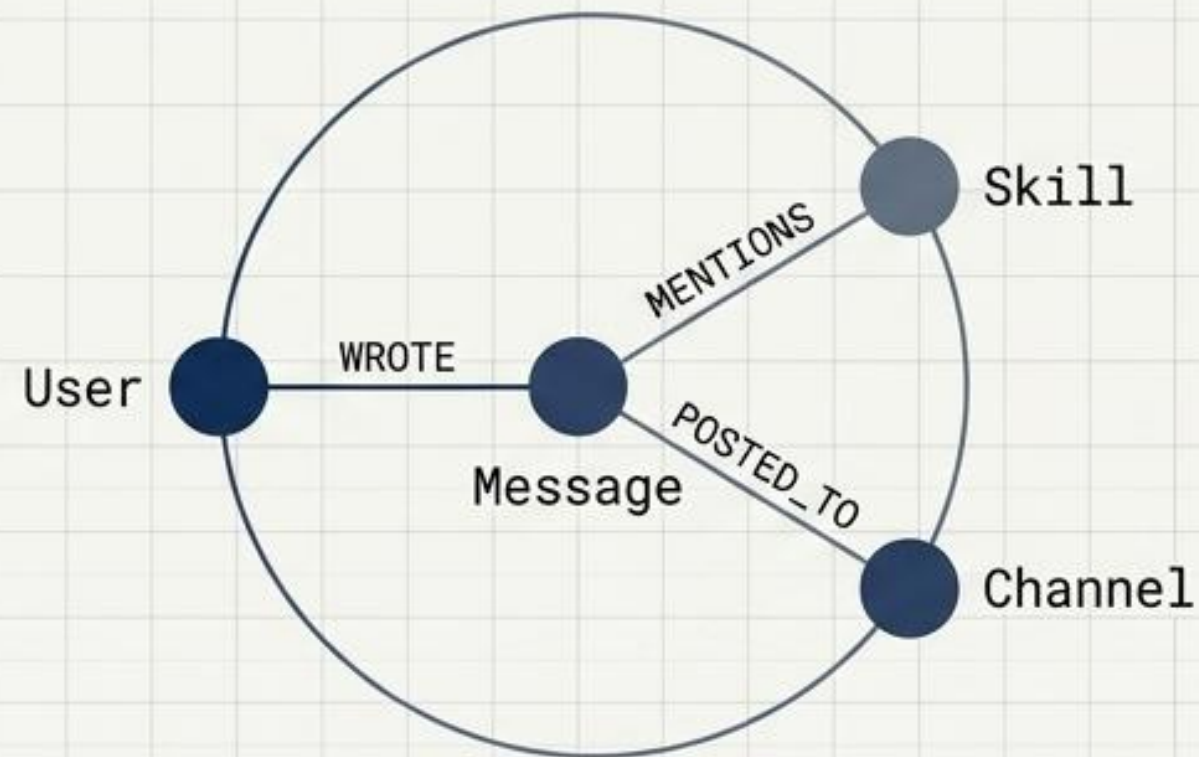
$[Project\ Duration] \times [Recency] \times [Peer\ Collaboration] = True\ Skill\ Level$



Insight

An individual's proficiency is not what they self-report on an HR checklist. It is the aggregated mathematical sum of their actual exposure to technologies across verified projects over time.

Uncovering Hidden Champions



The Communication Graph

Ingesting unstructured data from platforms like Slack or Teams transforms chaotic chat logs into structured expertise mappings. The graph tracks thread depth, reply frequency, and technical topic mentions.

Knowledge Power over Org Power

This topology bypasses the formal HR org chart entirely. It automatically surfaces the true, decentralized experts who hold the "knowledge power" and support the community, regardless of their official job title.

Building Elite Teams via Topological Chemistry

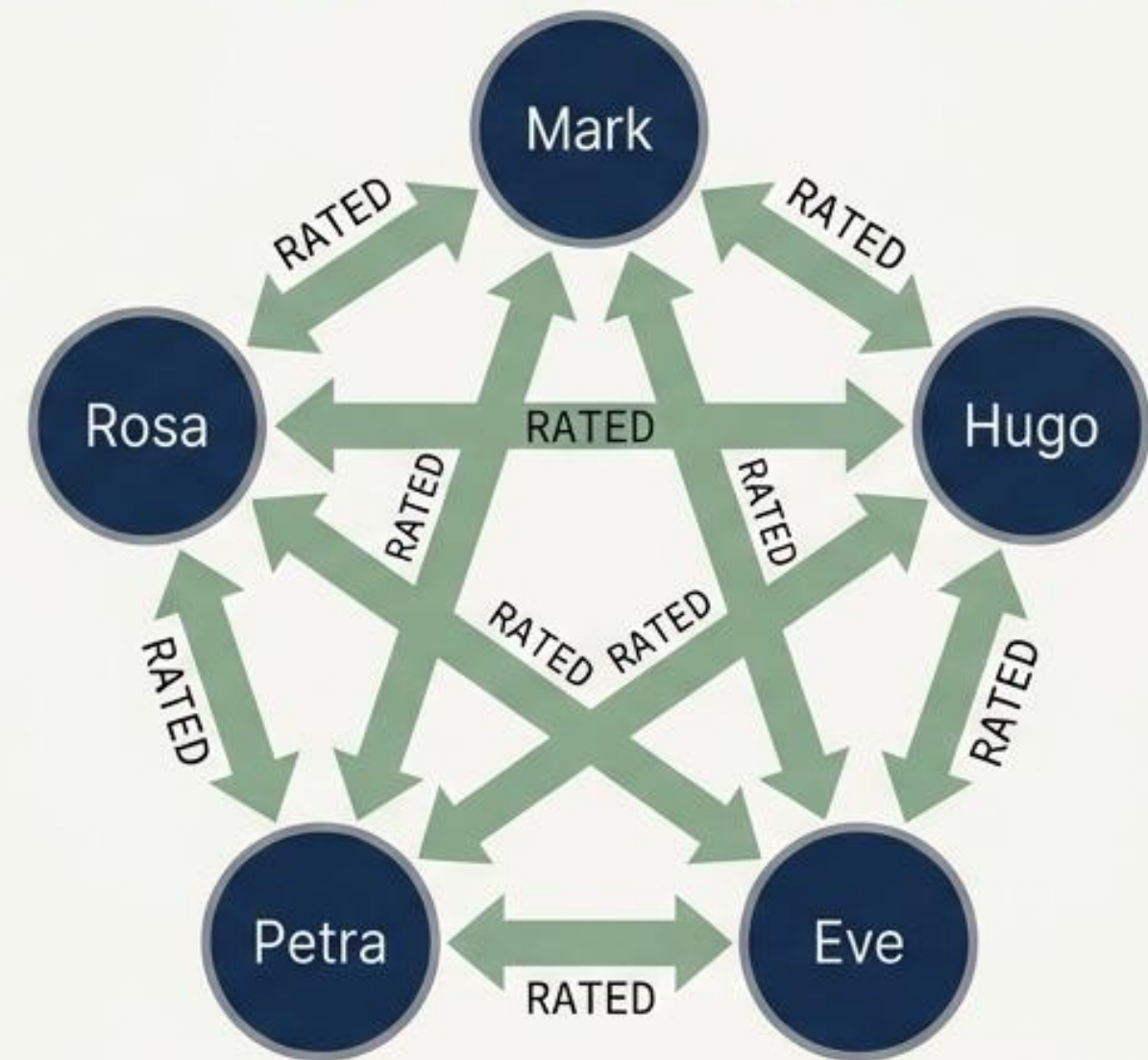
1. The Social Layer

Technical prerequisites are insufficient for high performance. We inject social compatibility topologies—historical peer ratings from past projects—into the team assembly algorithm.

2. The Chemistry Output

The graph generates precise recommendations for new project pods based on proven historical cohesion. Selecting subgraphs with mutual high ratings ($\text{Score} \geq 5$) drastically reduces friction and accelerates time-to-value for newly formed units.

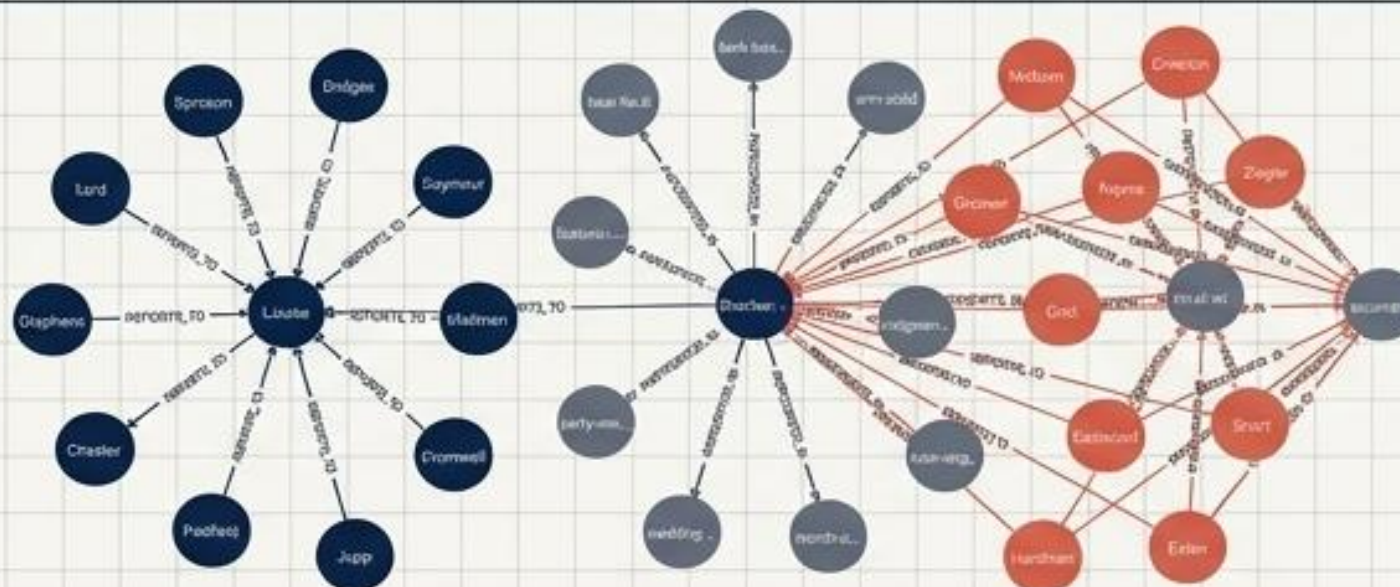
Peer-Rating Subgraph



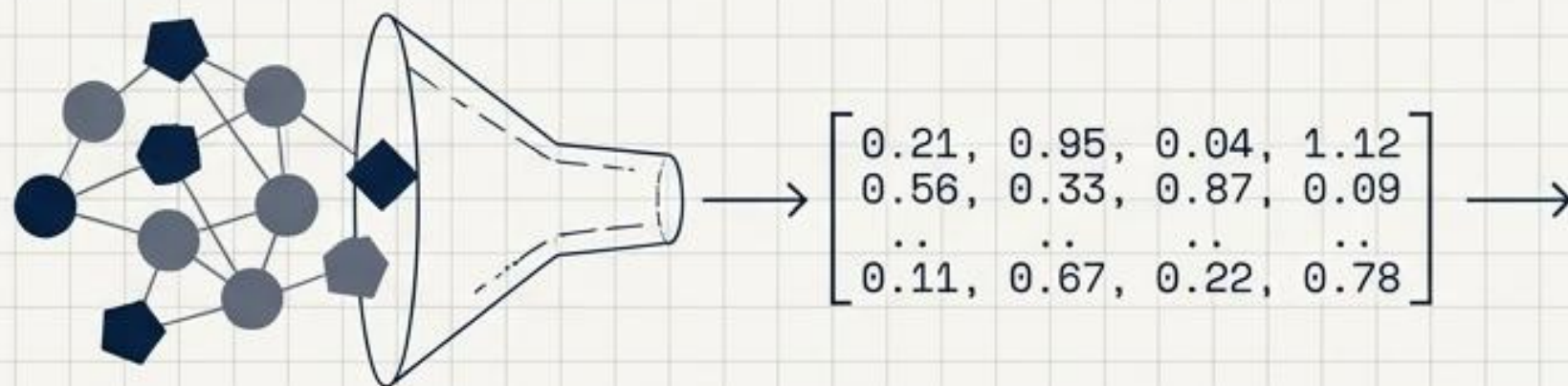
Predicting Project Outcomes with Machine Learning

1. Extract Topology

Map the human capital topology encompassing managers, contributors, and active projects.

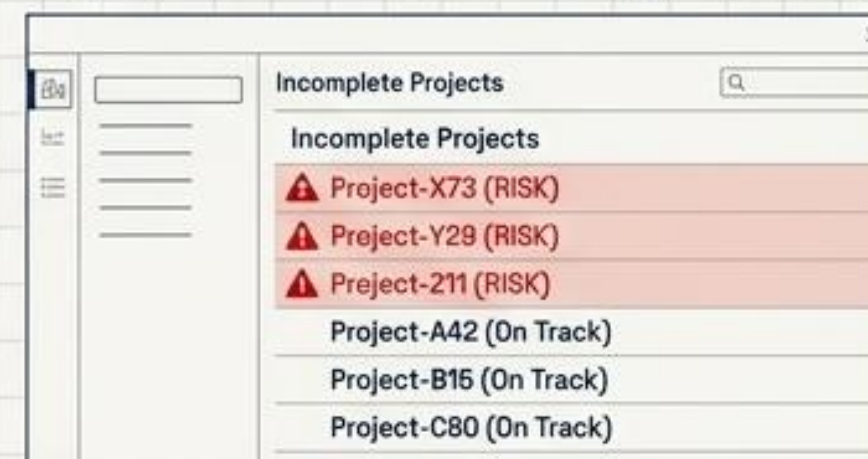


2. Embed as Features (FastRP)



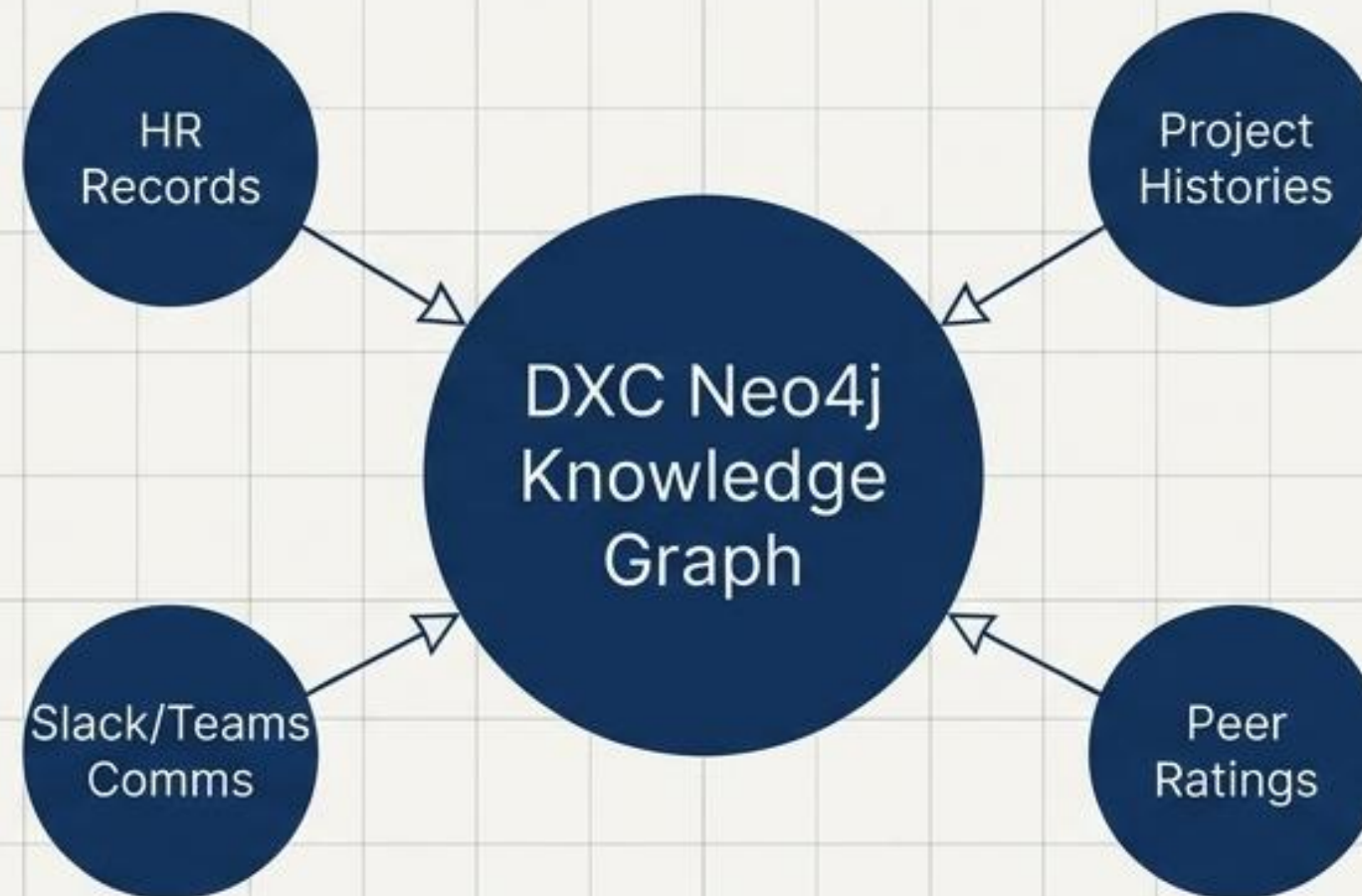
Use FastRP algorithms to transform complex node relationships and organizational topologies into numeric feature vectors.

3. Predict Outcome (Random Forest Regression)



Deploy regression models to flag in-flight projects exhibiting the exact topological markers of historical failure.

Case Study: DXC Technology Career Navigator



The Challenge

Managing 130,000 employees globally, DXC was restricted by siloed HR data resulting in major blind spots for internal mobility, cross-skilling, and optimal deployment.

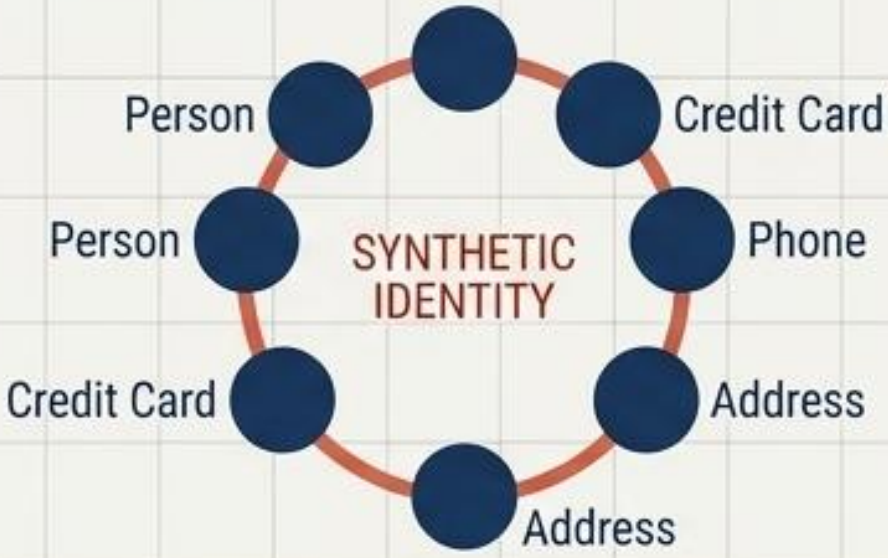
The Solution

Ingested all isolated employee data sources into a unified Neo4j Knowledge Graph, creating a living structural view of every employee and their latent, historically proven skills.

The Impact

The Career Navigator app now maps precise, graph-traversal skill-gap pathways between an employee's current state and high-value internal roles, boosting retention and strategic readiness.

The Universal Value of Pattern Detection

	Domain 1: Fraud Detection	Domain 2: Skills Matching
Core Objective	Defensive Posture (Protect Assets)	Offensive Posture (Maximize Human Capital)
Key Entities Mapped	People, Credit Cards, Phones, Addresses	People, Projects, Skills, Messages
Hidden Truth Discovered	The Synthetic Identity Ring 	The True Organizational Chart 
Algorithms Applied	Community Detection (Louvain)	Feature Embedding (FastRP) & Regression



Tabular data shows you what happened. Pattern Detection Knowledge Graphs show you how reality is connected—and what will happen next.

